EXPERIMENT-01

SIMULATION-BASED ELECTROMAGNETIC ANALYSIS OF TEM, TE AND TM PROPAGATION MODES IN MICROWAVE TRANSMISSION STRUCTURES

PROGRAM

```
clc;

clear;

close all;

%% CONSTANTS

c = 3e8;

mu0 = 4*pi*1e-7;

eps0 = 8.854e-12;

mu = mu0;

eps = eps0;

v = 1/sqrt(mu*eps);

eta = sqrt(mu/eps);

%% RECTANGULAR WAVEGUIDE

a = 22.86e-3;

b = 10.16e-3;

f = 10e9;

w = 2*pi*f;

k = w/v;

lambda = v/f;


%% CUTOFF FREQUENCIES
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fc_TE10 = c/(2*a);
fc_TE20 = c/a;
fc_TE01 = c/(2*b);
fc_TM11 = (c/2)*sqrt((1/a)^2+(1/b)^2);
fc = [fc_TE10 fc_TE20 fc_TE01 fc_TM11];
names = {'TE10','TE20','TE01','TM11'};
fprintf('\n--- MICROWAVE MODE ANALYSIS ---\n');
fprintf('Operating frequency = %.2f GHz\n',f/1e9);
fprintf('Wavelength = %.2f mm\n',lambda*1e3);
fprintf('Wave impedance = %.2f Ohms\n',eta);
fprintf('\nCutoff Frequencies:\n');
for i = 1:length(fc)
    fprintf('%s = %.3f GHz\n',names{i},fc(i)/1e9);
end
%% PROPAGATION CONSTANTS
fprintf('\nPropagation Analysis:\n');
for i = 1:length(fc)
    if f > fc(i)
        beta = k*sqrt(1-(fc(i)/f)^2);
        fprintf('%s: Propagating, beta = %.3f rad/m\n',...
            names{i},beta);
    else
        alpha = k*sqrt((fc(i)/f)^2-1);
        fprintf('%s: Below cutoff, alpha = %.3f Np/m\n',...
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        names{i},alpha);
    end
end
%% FIELD CALCULATION GRID
Nx = 60;
Ny = 40;
x = linspace(0,a,Nx);
y = linspace(0,b,Ny);
[X,Y] = meshgrid(x,y);
%% TE10 FIELDS
Ey = sin(pi*X/a);
Hz = cos(pi*X/a);
%% TM11 FIELDS
Ez = sin(pi*X/a).*sin(pi*Y/b);
Ex_TM = (pi/a)*cos(pi*X/a).*sin(pi*Y/b);
Ey_TM = (pi/b)*sin(pi*X/a).*cos(pi*Y/b);
%% TEM FIELDS
d = 10e-3;
W = 30e-3;
V0 = 1;
xt = linspace(0,d,40);
yt = linspace(0,W,40);
[XT,YT] = meshgrid(xt,yt);
E0 = V0/d;
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ETEM = E0*ones(size(XT));
HTEM = ETEM/eta;
%% FREQUENCY RANGE
freq = linspace(1e9,20e9,500);
%% COMBINED SUBPLOT FIGURE
figure('Name','Microwave Mode Analysis',...
    'NumberTitle','off',...
    'Color','w');

% 1. TE10 Electric Field
subplot(3,3,1);
surf(X*1000,Y*1000,Ey,'EdgeColor','none');
title('TE10 Electric Field');
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('E_y');
colorbar;
view(45,30);
grid on;
% 2. TE10 Magnetic Field
subplot(3,3,2);
surf(X*1000,Y*1000,Hz,'EdgeColor','none');
title('TE10 Magnetic Field');
xlabel('x (mm)');
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ylabel('y (mm)');
zlabel('H_z');
colorbar;
view(45,30);
grid on;
% 3. TM11 Longitudinal Electric Field
subplot(3,3,3);
surf(X*1000,Y*1000,Ez,'EdgeColor','none');
title('TM11 Longitudinal E Field');
xlabel('x (mm)');
ylabel('y (mm)');
zlabel('E_z');
colorbar;
view(45,30);
grid on;
% 4. TM11 Transverse Electric Field
subplot(3,3,4);
quiver(X*1000,Y*1000,Ex_TM,Ey_TM);
title('TM11 Transverse E Field');
xlabel('x (mm)');
ylabel('y (mm)');
axis equal tight;
grid on;
% 5. TEM Electric Field
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subplot(3,3,5);
surf(XT*1000,YT*1000,ETEM,'EdgeColor','none');
title('TEM Electric Field');
xlabel('Separation (mm)');
ylabel('Width (mm)');
zlabel('E (V/m)');
colorbar;
view(45,30);
grid on;
% 6. TEM Magnetic Field
subplot(3,3,6);
surf(XT*1000,YT*1000,HTEM,'EdgeColor','none');
title('TEM Magnetic Field');
xlabel('Separation (mm)');
ylabel('Width (mm)');
zlabel('H (A/m)');
colorbar;
view(45,30);
grid on;
% 7. Cutoff Frequency Comparison
subplot(3,3,7);
bar(fc/1e9);
set(gca,'XTickLabel',names);
ylabel('Frequency (GHz)');
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```
title('Cutoff Frequencies');
hold on;
yline(f/1e9,'r--','Operating Frequency');
grid on;
hold off;
% 8. Propagation Constant vs Frequency
subplot(3,3,8);
hold on;
for i = 1:length(fc)
    beta = nan(size(freq));
    idx = freq > fc(i);
    beta(idx) = (2*pi*freq(idx)/v).* ...
        sqrt(1-(fc(i)./freq(idx)).^2);
    plot(freq/1e9,beta,'LineWidth',1.2);
end
xlabel('Frequency (GHz)');
ylabel('\beta (rad/m)');
title('Propagation Constant');
legend(names,'Location','northwest');
grid on;
hold off;
% 9. TE and TM Field Comparison
subplot(3,3,9);
imagesc(x*1000,y*1000,Ey);
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set(gca,'YDir','normal');

xlabel('x (mm)');

ylabel('y (mm)');

title('TE10 Electric Field');

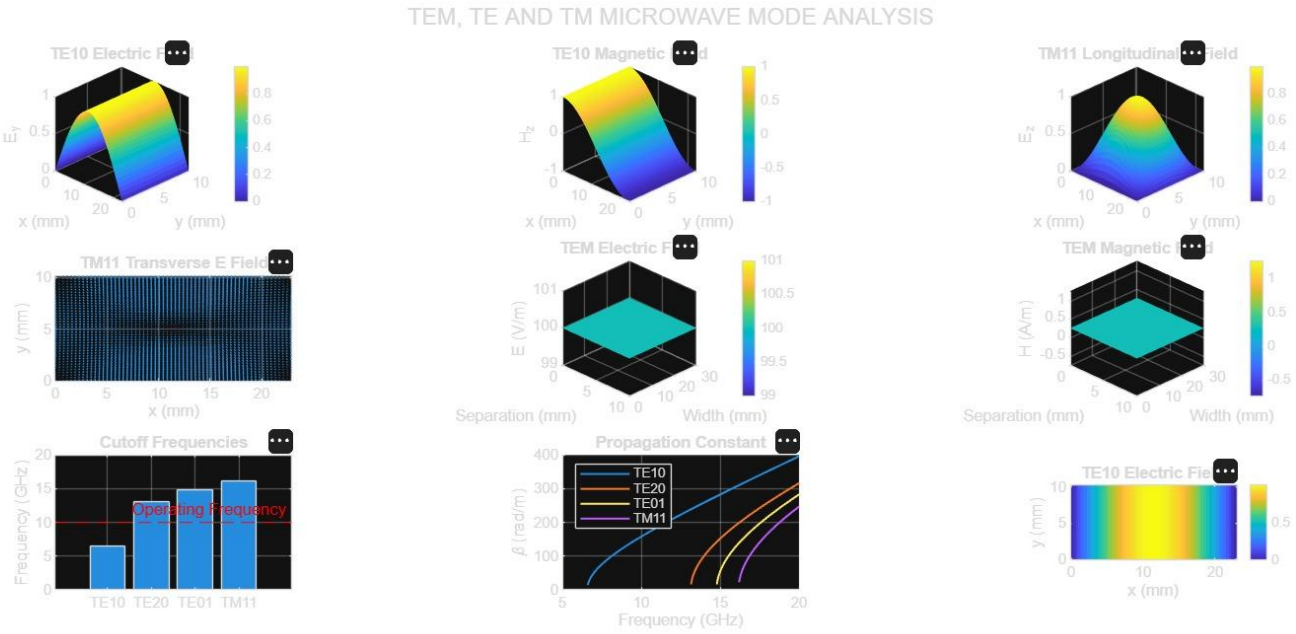
colorbar;

axis image;

sgtitle('TEM, TE AND TM MICROWAVE MODE ANALYSIS');

fprintf('\nSimulation completed successfully.\n');
```

OUTPUT



RESULT

The simulation confirms that TE_{10} propagates at 10 GHz, TM_{11} is below cutoff, and a hollow rectangular waveguide does not support TEM mode. The code generates five MATLAB figures.